

# A Modification of Floer Cohomology for Doubling Spaces

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Preprint available at [arXiv:2507.08181](https://arxiv.org/abs/2507.08181)  
(QR code links to abstract)

# String Theory and Geometry

Witten's dictionary<sup>1</sup> between algebraic geometry and string theory:

| Physics               | Algebraic Geometry               |
|-----------------------|----------------------------------|
| Spacetime Geometry    | projective $\mathbb{C}$ -variety |
| B-model supersymmetry | Complex structure                |
| A-model supersymmetry | Symplectic/Kähler structure      |
| B-model D-branes      | Derived category $D^b(X)$        |
| A-model D-branes      | Fukaya category $\text{Fuk}(X)$  |



<sup>1</sup>Witten arXiv:hep-th/9112056

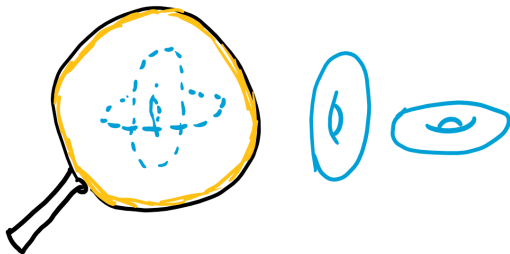
## Motivation for Doubling: T-duality

Most symmetries of a string theory are “geometric” i.e. come from geometric symmetries of the target space, but...

### Theorem (T-duality)

*String theory on a torus is isomorphic to string theory on the dual torus (via field redefinitions).*

What geometry does the string theory see?



# Doubling Spaces

## Definition

For  $X$  a torus, the *doubling space* for  $X$  is

$$\mathbb{X} = X \times \hat{X}$$

## Theorem

*String theory on  $X$  = String theory on  $\mathbb{X}$  plus constraints.<sup>a</sup>*

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<sup>a</sup>Siegel arXiv:hep-th/9305073 and hep-th/9302036, Hull and Zwiebach arXiv:0904.4664

## Question

*Can we extract algebraic geometry information about a manifold using doubling spaces?*

Symplectic side: yes, holomorphic Lagrangian intersection cohomology in  $\mathbb{X}$  encodes Fukaya category information<sup>2</sup>

Complex side: this talk

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<sup>2</sup>Y. Qin arXiv:2207.10647

# Caveat Emptor: We Don't Know Many Doubling Spaces

Any torus can be doubled

Homogeneous spaces can sometimes be doubled: Drinfel'd double structure...<sup>3</sup>

???

## Warning

*We will need to use results specific to tori for most of our results, but we will try to hint at a general theory when possible.*

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<sup>3</sup>Hull and Reid-Edwards:arxiv:0902.4032

# Local Geometry

What does local geometry of  $\mathbb{X}$  look like?

- symplectic structure from Poincaré bundle:

$$c_1(\mathcal{P}) = \sum dx_i \wedge d\hat{x}_i$$

- “foliation operator” from product structure:

$$F \in \text{End}(T\mathbb{X}), \quad F|_{TX} = 1, \quad F|_{T\hat{X}} = -1$$

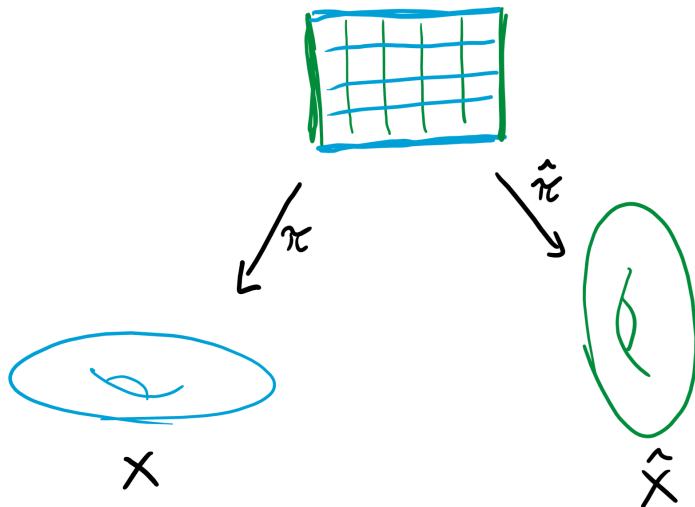
- canonical neutral metric from dual structure:

$$G(X, \omega) = \omega(X)$$

Proposition (two-out-of-three property)

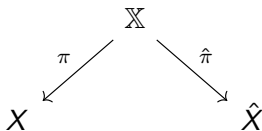
*Any two of these determine the third.*

# Local Geometry





# Geometry From the Base



## Theorem (Structure of the tangent space)

*There is a canonical isomorphism of smooth vector bundles*

$$T\tilde{X} \cong \pi^*(TX \oplus T^*X) \cong \hat{\pi}^*(T\hat{X} \oplus T^*\hat{X})$$

Proof idea:  $c_1(\mathcal{P})$  furnishes an isomorphism between the cotangent and normal bundles of the fibers.

# Geometry From the Base

Consequences of

$$T\mathbb{X} = \pi^* TX \oplus \pi^* T^*X$$

| Geometric structure | Lift to $\mathbb{X}$ |
|---------------------|----------------------|
|---------------------|----------------------|

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|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| Metric $g$ | Doubled metric $g \oplus g^{-1} = \begin{bmatrix} 0 & g^{-1} \\ g & 0 \end{bmatrix}$ |
|------------|--------------------------------------------------------------------------------------|

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| Complex structure $J$ | Doubled cx. structure $J \oplus -J^* = \begin{bmatrix} J & 0 \\ 0 & -J^* \end{bmatrix}$ |
|-----------------------|-----------------------------------------------------------------------------------------|

|                          |                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Symplectic form $\omega$ | Doubled symp. form $-\omega \oplus \omega^{-1} = \begin{bmatrix} 0 & \omega^{-1} \\ -\omega & 0 \end{bmatrix}$ |
|--------------------------|----------------------------------------------------------------------------------------------------------------|

# Unusual Geometry From the Base

Doubling a  $B$ -field yields a metric:  $B \in H^2(X, \mathbb{R})$ ,

$$(g + B) \mapsto G^B = \begin{bmatrix} -g^{-1}B & g^{-1} \\ g - Bg^{-1}B & Bg^{-1} \end{bmatrix}$$

a (positive-definite) metric on  $\mathbb{X}$ .

# Unusual Geometry From the Base

Doubling  $\omega$  or  $J$  yields *complex* structures:

$$\begin{bmatrix} J & 0 \\ 0 & -J^* \end{bmatrix}^2 = \begin{bmatrix} J^2 & 0 \\ 0 & (-J^*)^2 \end{bmatrix} = -I$$

$$\begin{bmatrix} 0 & \omega^{-1} \\ -\omega & 0 \end{bmatrix}^2 = \begin{bmatrix} -\omega^{-1}\omega & 0 \\ 0 & -\omega\omega^{-1} \end{bmatrix} = -I$$

# Generalized Complex Geometry

Generalized complex geometry is a framework that unifies complex and symplectic geometry.

## Definition

A *generalized complex structure* on  $X$  is an endomorphism  $\mathcal{J}$  of  $TX \oplus T^*X$  which satisfies  $\mathcal{J}^2 = -1$  and  $\mathcal{J}^* = -\mathcal{J}$  (under the canonical evaluation metric on  $TX \oplus TX^*$ ).

Examples:

$$\mathcal{J}_J = \begin{bmatrix} J & 0 \\ 0 & -J^* \end{bmatrix} \qquad \mathcal{J}_\omega = \begin{bmatrix} 0 & \omega^{-1} \\ -\omega & 0 \end{bmatrix}$$

# Doubling a GC Structure

## Theorem

*Under the isomorphism  $T\mathbb{X} \cong \pi^*(TX \oplus T^*X)$ , GC structures on  $X$  pull back to (almost) complex structures on  $\mathbb{X}$*

$B$ -fields lift to metrics

GC structures lift to complex structures

$\mathbb{X}$  “geometrizes” these non-geometric structures!

# The Problem

| Physics          | Algebraic Geometry              | Doubled Geometry                |
|------------------|---------------------------------|---------------------------------|
| Torus geometry   | Complex torus $X$               | $\mathbb{X} = X \times \hat{X}$ |
| B-model SUSY     | Complex structure               | $\mathcal{I}_J$                 |
| A-model SUSY     | Symplectic structure            | $\mathcal{I}_\omega$            |
| General SUSY     | GC structure                    | $\mathcal{I}$                   |
| B-model D-branes | Derived category $D^b(X)$       | ???                             |
| A-model D-branes | Fukaya category $\text{Fuk}(X)$ | ???                             |

**A-model:** How do we lift Lagrangian subtori?

**B-model:** How do we lift line bundles?

## Hint: Generalized Tangent Spaces

### Definition

Let  $(M, \nabla)$  be a submanifold of  $X$  along with a  $U(1)$  connection on  $M$ . Then, the *generalized tangent bundle* of  $(M, \nabla)$  is the subbundle of  $TX \oplus T^*X|_M$  defined by

$$\mathcal{T}M := \{X + \xi \mid X \in TM, \xi|_M = \iota_X F\}$$

with  $F = \nabla^2$  the curvature of the connection.

A pair  $(M, \nabla)$  is a *generalized complex submanifold* if  $\mathcal{T}M$  is stable under  $\mathcal{J}$ .



# Examples of GC Submanifolds

| GC Structure         | GC Submanifolds                                                                         |
|----------------------|-----------------------------------------------------------------------------------------|
| $\mathcal{I}_J$      | Holomorphic submanifold +<br>holomorphic $U(1)$ connection                              |
| $\mathcal{I}_\omega$ | Lagrangian submanifold +<br>flat $U(1)$ connection<br>or coisotropic brane <sup>4</sup> |

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<sup>4</sup>In the sense of Kapustin-Orlov arXiv:hep-th/0109098  
and Aldi-Zaslow arXiv:hep-th/0501247

# The Problem

For  $(M, \nabla)$  a GC submanifold, we seek a geometric object

$$\mathbb{M} \subseteq \mathbb{X}$$

We know what the tangent space should be:

$$\mathcal{T}M = \{X + \xi \mid X \in TM, \xi|_M = \iota_X F|_M\}$$

$$T\mathbb{M} \stackrel{?}{=} \pi^* \mathcal{T}M$$

# The Doubled A-Model

Previous work by Y. Qin <sup>1</sup>

## Definition

The *lift* of an A-brane (GC submanifold for  $\mathcal{J}_\omega$ )  $(M, \nabla)$  is

$$\mathbb{M} = \{(x, \hat{x}) \mid x \in M, \hat{x}(\gamma_x) = (-1)^{\xi(\gamma_x)} \text{hol}_\nabla(\gamma_x) \forall \gamma_x \in \pi_1(M, x)\}$$

where  $\xi$  is a  $\mathbb{Z}_2$  semi-character for  $F_\nabla$ .

$x \in M$ :  $\mathbb{M}$  covers  $M$

$\hat{x}$  condition: holonomy of  $\nabla$  is tracked by  $\hat{x}$

$(-1)^\xi$ : Untwists  $\text{hol}_\nabla$  to become a group homomorphism

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<sup>1</sup>Y. Qin arXiv:2207.10647

# The Doubled A-Model: Geometrization

Theorem (c.f. Y. Qin Prop. 3.6)

$$T\mathbb{M} = \pi^* \mathcal{T}M$$

and thus  $T\mathbb{M}$  is  $\mathcal{J}_\omega$ -invariant i.e.  $\mathcal{J}_\omega$ -holomorphic.

Theorem (Y. Qin, Thm. 1.3)

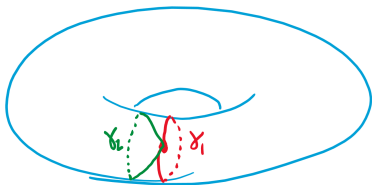
*There is an isomorphism between the Lagrangian Intersection Floer cohomology of two Lagrangian A-branes on  $X$  and a certain subgroup (“u-part”) of the Floer cohomology of their lifts.*

*Furthermore, the “u-part” Floer cohomology is defined for coisotropic branes.*

# Doubling the B-Model

We focus now on a complex torus  $X$  and space-filling B-branes, which are simply holomorphic line bundles on  $X$ .

Curvature  $F$  is no longer zero or  $B$ -field shift of zero.



$$\text{If } F \neq 0 \\ \text{hol}_\nabla(\gamma_1) \neq \text{hol}_\nabla(\gamma_2)$$



$$\text{hol}_\nabla(\gamma_1) - \text{hol}_\nabla(\gamma_2) \\ = \int F$$

# Lifting B-Branes: $E$ -Flatness

## Definition

A connection  $\nabla$  on a line bundle  $\mathcal{L}$  is  $E$ -flat if  $c_1(\mathcal{L}) = E$  and  $F_\nabla$  is translation-invariant.

Flat =  $E$ -flat with  $E = 0$

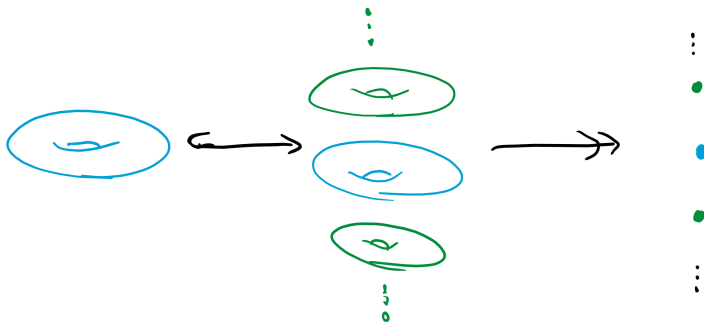
## Proposition (every line bundle has an $E$ -flat representative)

*There is a canonical isomorphism between  $\text{Pic}^E(X) := c_1^{-1}(E)$  and the space of  $E$ -flat connections.*

Since  $E$ -flat = harmonic in flat metric, an  $E$ -flat connection is a special case of a Yang-Mills connection.

# E-Flatness

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathrm{Pic}^0(X) & \longrightarrow & \mathrm{Pic}(X) & \xrightarrow{c_1} & H^{(1,1)}(X, \mathbb{Z}) \longrightarrow 0 \\
 & & \downarrow \cong & & \downarrow \cong & & \downarrow = \\
 0 & \longrightarrow & \mathcal{M}_{\mathrm{flat}} & \longrightarrow & \bigoplus_E \mathcal{M}_{E\text{-flat}} & \longrightarrow & H^{(1,1)}(X, \mathbb{Z}) \longrightarrow 0
 \end{array}$$



# Lifting B-Branes: Symmetric Semi-Characters

## Definition

A *symmetric semi-character* for  $E$  is a map  $\xi : H_1(X, \mathbb{Z}) \rightarrow \mathbb{Z}_2$  satisfying

$$\xi(\gamma + \gamma') = \xi(\gamma) + \xi(\gamma') + E(\gamma \wedge \gamma')$$

## Proposition

*Symmetric semi-characters exist for any  $E$ , and the space of such semi-characters is affine over  $H^1(X, \mathbb{Z}_2)$ .*

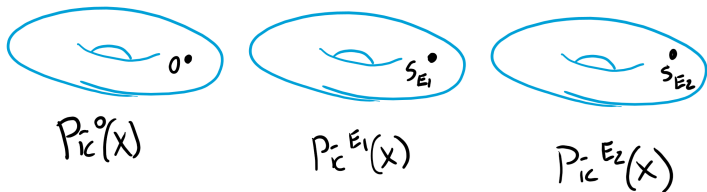
## Proposition (H.)

*If  $E \in H^2(X, \mathbb{Z})$  is an integral translation-invariant  $(1, 1)$ -form, then for each symmetric semi-character  $\xi$  for  $E$  there exists a unique  $E$ -flat holomorphic line bundle  $S_{E, \xi}$  whose holonomy is given by  $(-1)^\xi$ .*



# Lifting B-Branes: Symmetric Semi-Characters

Choose a system of such line bundles  $S_E$  for each  $E \in H^{(1,1)}(X, \mathbb{Z})$  with  $S_{E_1} \otimes S_{E_2} \cong S_{E_1+E_2}$  (“normalizing factors”)



The  $S_E$  serve as “identity elements” for the Picard groups.

# Lifting B-Branes

## Definition

For  $\mathcal{L}$  a holomorphic line bundle on  $X$  with  $c_1(\mathcal{L}) = E$ , the lift of  $\mathcal{L}$ , denoted  $\mathbb{L}(\mathcal{L})$  is the graph

$$\mathbb{L}(\mathcal{L}) = \Gamma(x \mapsto (S_{-E} \otimes t_{-x}^* \mathcal{L}) \in \text{Pic}^0(X))$$

$S_{-E}$ : line bundle version of  $(-1)^\xi$

$t_{-x}^* \mathcal{L}$ : line bundle version of  $\text{hol}_\nabla(\gamma + x)$

# Connections to Generalized Complex Geometry

## Theorem (H.)

*This definition extends Y. Qin's definition of the lift to holomorphic  $U(1)$  connections. It can be further extended to arbitrary GC submanifolds.*

## Proposition (H., similar to Y. Qin Prop. 3.6)

*Let  $c_1(\mathcal{L}) = E$ . Then, under the isomorphism  $T\mathbb{X} \cong \pi^*(TX \oplus T^*X)$ ,*

$$T\mathbb{L}(\mathcal{L}) = \pi^*\mathcal{T}_E X$$

*where  $\mathcal{T}_E X$  is the generalized tangent bundle of  $(X, E)$ .*

## Corollary

*$\mathbb{L}(\mathcal{L})$  is a maximal isotropic submanifold under the canonical neutral metric on  $\mathbb{X}$ . Furthermore, it is  $\mathcal{J}_J$ -holomorphic and  $\mathcal{J}_J^\sharp$ -Lagrangian.*

## Examples

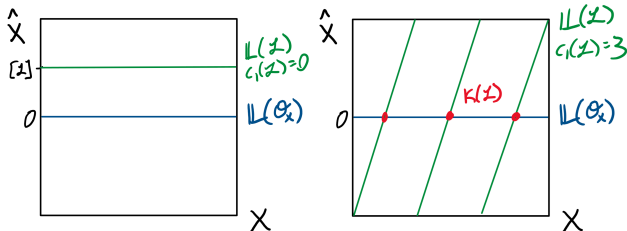
For  $\mathcal{O}_X$  the structure sheaf,

$$\mathbb{O}_X := \mathbb{L}(\mathcal{O}_X) = \Gamma(x \mapsto S_0 \cong \mathcal{O}_X) = X \times \{0\}$$

If  $c_1(\mathcal{L}) = 0$  then  $t_x^* \mathcal{L} = \mathcal{L}$  and

$$\mathbb{L}(\mathcal{L}) = \Gamma(x \mapsto S_0 \otimes \mathcal{L}) = X \times \{[\mathcal{L}]\}$$

Otherwise,  $c_1(\mathcal{L})$  is a slope.



# Properties of the Lift

## Proposition (H.)

*The lifting map  $\mathbb{L} : \text{Pic}(X) \rightarrow \mathcal{M}_L(\mathbb{X})$  is  $\text{Pic}^0$ -equivariant.*

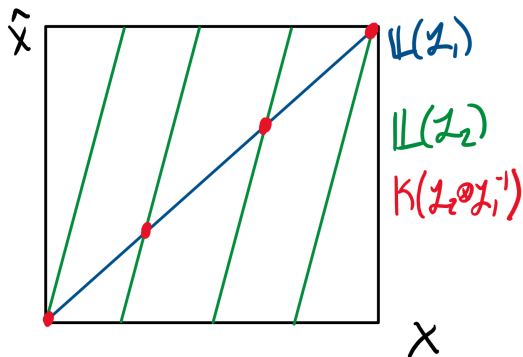
## Proposition (H.)

*For  $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L} \in \text{Pic}(X)$ ,*

$$\mathbb{L}(\mathcal{L}_1) \cap \mathbb{L}(\mathcal{L}_2) = t_{\hat{x}}(\mathbb{L}(\mathcal{L}_1 \otimes \mathcal{L}) \cap \mathbb{L}(\mathcal{L}_2 \otimes \mathcal{L}))$$

*for some  $\hat{x} \in \hat{X}$ .*

# Properties of the Lift



# Intersections of Lifts

Using the invariance of the lift under tensors, we find

$$\mathbb{L}(\mathcal{L}_1) \cap \mathbb{L}(\mathcal{L}_2) = t_{\hat{x}}(\mathbb{O}_X \cap \mathbb{L}(\mathcal{L}_2 \otimes \mathcal{L}_1^{-1}))$$

Arbitrary intersections can be computed:

## Theorem (H.)

Let  $c_1(\mathcal{L}_1) = E_1$  and  $c_1(\mathcal{L}_2) = E_2$ , and define

$$\begin{aligned}\phi : X &\rightarrow \hat{X} \\ x &\mapsto t_x^* S_{E_1 - E_2} \otimes S_{E_1 - E_2}^{-1}\end{aligned}$$

Then the intersection of  $\mathbb{L}(\mathcal{L}_1)$  and  $\mathbb{L}(\mathcal{L}_2)$  is either empty or (up to translations in  $X, \hat{X}$ )

$$\mathbb{L}(\mathcal{L}_1) \cap \mathbb{L}(\mathcal{L}_2) = \ker(\phi(x))$$

# Self-Intersections and the Modified Lagrangian Floer Cohomology

Fact from Lagrangian Floer theory:

## Theorem

*In sufficiently nice cases (e.g. if  $L$  is a Lagrangian subtorus of a symplectic torus)*

$$HF^*(L, L) \cong H^*(L, \mathbb{C})$$

Idea of proof: perturb  $L$  by the normal vector field generated by  $df$  for suitably nice  $f : L \rightarrow \mathbb{R}$ , the Floer theory is now computable using Morse theory on the Morse function  $f$ .



# Self-Intersections and the Modified Lagrangian Floer Cohomology

How do we recover  $\text{Ext}^*(\mathcal{L}, \mathcal{L})$  from  $HF^*(\mathbb{L}(\mathcal{L}), \mathbb{L}(\mathcal{L}))$ ?

$$HF^*(\mathbb{L}(\mathcal{L}), \mathbb{L}(\mathcal{L})) = H^*(X, \mathbb{C})$$

is too big, but  $\mathcal{J}$  induces a Hodge decomposition.

## Definition

The  $\mathcal{J}$ -holomorphic part of  $HF^*(\mathbb{L}(\mathcal{L}), \mathbb{L}(\mathcal{L})) \cong H^*(\mathbb{L}(\mathcal{L}), \mathbb{C})$  is the  $(0, *)$  part under the Hodge decomposition defined by  $\mathcal{J}$ .

$$HF_{\mathcal{J}}^*(\mathbb{L}(\mathcal{L}), \mathbb{L}(\mathcal{L})) = H^{0,*}(\mathbb{L}(\mathcal{L}), \mathbb{C})$$

# Connection to Physics

## Theorem (H.)

Suppose  $c_1(\mathcal{L}_1) = c_1(\mathcal{L}_2)$ . Then,

$$\mathrm{Hom}_B(\mathcal{L}_1, \mathcal{L}_2) := \bigoplus_q \mathrm{Ext}^q(\mathcal{L}_1, \mathcal{L}_2) = \bigoplus_q HF_{\mathcal{J}}^q(\mathbb{L}(\mathcal{L}_1), \mathbb{L}(\mathcal{L}_2))$$

This is either empty if  $\mathcal{L}_1 \neq \mathcal{L}_2$  or  $H^*(X, \mathcal{O}_X)$  if  $\mathcal{L}_1 = \mathcal{L}_2$ .

## What about other line bundles?

Suppose  $\mathcal{L}$  is “positive-definite”, then its cohomology is concentrated in degree zero.

$$\mathrm{Hom}_B(\mathcal{O}_X, \mathcal{L}) = \mathrm{Hom}^0(\mathcal{O}_X, \mathcal{L}) = H^0(\mathcal{L})$$

But,

$$\mathbb{L}(\mathcal{L}) \cap \mathbb{O}_X = K(\mathcal{L})$$

and

$$\#(K(\mathcal{L})) = (h^0(\mathcal{L}))^2$$

So, the dimension of the Floer complex is off by a square.

# Possible Resolution

$K(\mathcal{L})$  comes with a “canonical” decomposition

$$K(\mathcal{L}) \cong \Lambda_{\mathcal{L}}^{\oplus 2}$$

So,  $HF_{\mathcal{J}}^q$  in this case may have to be defined by averaging over one of the factors.

The same structure, however, is found on the A-model side. There, an averaging over half of a lattice was necessary in the case of transverse intersections.

# Summary

$\mathbb{X} = X \times \hat{X}$  geometrizes some non-geometric constructions on  $X$

- B-field becomes metric twist
- GC structures become complex structures

A-type branes (nice ones) lift to  $\mathcal{I}_\omega$ -holomorphic,  $\mathcal{I}_\omega^\sharp$ -Lagrangian subtori.

B-type branes (nice ones) lift to  $\mathcal{I}_J$ -holomorphic,  $\mathcal{I}_J^\sharp$ -Lagrangian subtori.

String-theoretic hom-spaces (Fukaya/derived) can be computed (in nice cases) by “holomorphic” Lagrangian Intersection Floer cohomology.

Thank you!